

| Assessment | Test | Domain        | Skill               | Difficulty                                          |
|------------|------|---------------|---------------------|-----------------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear functions | Medium <div><div></div><div></div><div></div></div> |

Question

| $x$  | $f(x)$ |
|------|--------|
| $-1$ | $10$   |
| $0$  | $14$   |
| $1$  | $20$   |

For the quadratic function  $f$ , the table shows three values of  $x$  and their corresponding values of  $f(x)$ . Which equation defines  $f$ ?

Answer

- A.  $f(x) = 3x^2 + 3x + 14$
- B.  $f(x) = 5x^2 + x + 14$
- C.  $f(x) = 9x^2 - x + 14$
- D.  $f(x) = x^2 + 5x + 14$

Correct Answer: D

Rationale

Choice D is correct. The equation of a quadratic function can be written in the form  $f(x) = a(x - h)^2 + k$ , where  $a$ ,  $h$ , and  $k$  are constants. It's given in the table that when  $x = -1$ , the corresponding value of  $f(x)$  is  $10$ . Substituting  $-1$  for  $x$  and  $10$  for  $f(x)$  in the equation  $f(x) = a(x - h)^2 + k$  gives  $10 = a(-1 - h)^2 + k$ , which is equivalent to  $10 = a(1 + 2h + h^2) + k$ , or  $10 = a + 2ah + ah^2 + k$ . It's given in the table that when  $x = 0$ , the corresponding value of  $f(x)$  is  $14$ . Substituting  $0$  for  $x$  and  $14$  for  $f(x)$  in the equation  $f(x) = a(x - h)^2 + k$  gives  $14 = a(0 - h)^2 + k$ , or  $14 = ah^2 + k$ . It's given in the table that when  $x = 1$ , the corresponding value of  $f(x)$  is  $20$ . Substituting  $1$  for  $x$  and  $20$  for  $f(x)$  in the equation  $f(x) = a(x - h)^2 + k$  gives  $20 = a(1 - h)^2 + k$ , which is equivalent to  $20 = a(1 - 2h + h^2) + k$ , or  $20 = a - 2ah + ah^2 + k$ . Adding  $20 = a - 2ah + ah^2 + k$  to the equation  $10 = a + 2ah + ah^2 + k$  gives  $30 = 2a + 2ah^2 + 2k$ . Dividing both sides of this equation by  $2$  gives  $15 = a + ah^2 + k$ . Since  $14 = ah^2 + k$ , substituting  $14$  for  $ah^2 + k$  into the equation  $15 = a + ah^2 + k$  gives  $15 = a + 14$ . Subtracting  $14$  from both sides of this equation gives  $a = 1$ . Substituting  $1$  for  $a$  in the equations  $14 = ah^2 + k$  and  $20 = ah^2 - 2ah + a + k$  gives  $14 = h^2 + k$  and  $20 = 1 - 2h + h^2 + k$ , respectively. Since  $14 = h^2 + k$ , substituting  $14$  for  $h^2 + k$  in the equation  $20 = 1 - 2h + h^2 + k$  gives  $20 = 1 - 2h + 14$ , or  $20 = 15 - 2h$ . Subtracting  $15$  from both sides of this equation gives  $5 = -2h$ . Dividing both sides of this equation by  $-2$  gives  $-\frac{5}{2} = h$ . Substituting  $-\frac{5}{2}$  for  $h$  into the equation  $14 = h^2 + k$  gives  $14 = (-\frac{5}{2})^2 + k$ , or  $14 = \frac{25}{4} + k$ . Subtracting  $\frac{25}{4}$  from both sides of this equation gives  $\frac{31}{4} = k$ . Substituting  $1$  for  $a$ ,  $-\frac{5}{2}$  for  $h$ , and  $\frac{31}{4}$  for  $k$  in the equation  $f(x) = a(x - h)^2 + k$  gives  $f(x) = (x + \frac{5}{2})^2 + \frac{31}{4}$ , which is equivalent to  $f(x) = x^2 + 5x + \frac{25}{4} + \frac{31}{4}$ , or  $f(x) = x^2 + 5x + 14$ . Therefore,  $f(x) = x^2 + 5x + 14$  defines  $f$ .

Choice A is incorrect. If  $f(x) = 3x^2 + 3x + 14$ , then when  $x = -1$ , the corresponding value of  $f(x)$  is  $14$ , not  $10$ .

Choice B is incorrect. If  $f(x) = 5x^2 + x + 14$ , then when  $x = -1$ , the corresponding value of  $f(x)$  is  $18$ , not  $10$ .

Choice C is incorrect. If  $f(x) = 9x^2 - x + 14$ , then when  $x = -1$ , the corresponding value of  $f(x)$  is  $24$ , not  $10$ , and when  $x = 1$ , the corresponding value of  $f(x)$  is  $22$ , not  $20$ .